HADHRAMOUT UNIVERSITY
COLLEGE OF COMPUTERS & INFORMATION TECHNOLOGY
Mid Term Exam

Academic Year : 2025 – 2026
Semester : The First
Department : IT
Level : The First

نموذج 1

Subject : *Differential Calculus*
Examiner : *Abeer AL Haddad*
Day and Date : Monday 17 / 11 / 2025
Time allowed : 1 hour

Question 1(10 marks)

Which of the following statements are correct (√) and which are incorrect (×)" (provide the reason)":

1- y – intercept of the line $\overleftrightarrow{L}: (3x - y + 8 = 0)$ is (4).

2- If $A(6\,,-2)\,, B(2,1)\ then$The Midpoint of $\overline{AB}$ = $\left(4\ , -\frac{1}{2}\right)$.

3- If $|x + 4| \leq 0\ \ then\ the\ inequality\ has\ no\ solution$.

4- The domain of the function $f(x) = \sqrt[3]{3x - 2}$ is $D_f = \boldsymbol{R}$.

5- The slope of the the line $\overleftrightarrow{L}$: ($y + x - 4 = 0$) is ($m = 4$).

Question 2 (10 marks)

a) Define the following :

1- The odd function

2- Distant formula d (A , B)

b) Let $f(x) = |x| + 5$, find the following :

1- $f(-3)$

2- $(D_f)\ domain\ f$

3- determine whether f is even , odd , neither even nor odd .

Question 3 (10 marks)

a) Find the following limits (if they exist):

1- $\lim_{x\to 6} \sqrt{2x - 3}$

2- $\lim_{x\to 0} \frac{\sqrt{1+x}-1}{x}$

b) Solve the following :

1- $|x - 2\,| - 1 = -4$

2- $\frac{x-1}{x-2} \leq 0$

c) Find the equation of the line that passes through the points $A(\,1\,,2)\,,\ B(3,2)$.

تمنياتي لكم بالتوفيق

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نموذج 2

Subject : *Differential Calculus*
Examiner : *Abeer AL Haddad*
Day and Date : Monday 17 / 11 / 2025
Time allowed : 1 hour

Question 1(10 marks)

Which of the following statements are correct (√) and which are incorrect (×)" (provide the reason)":

1- y – intercept of the line $\overleftrightarrow{L}: (2x - y + 5 = 0)$ is (5).
2- The distant between $A(4\,,-3)\ and\ B(2,1)$ $d(A\,,B) = \sqrt{17}$.
3- If $|x + 4| \leq -1$ $then\ the\ inequality\ has\ no\ solution$.
4- The domain of the function $f(x) = \sqrt{x-1}$ is $D_f = \boldsymbol{R}$.
5- The slope of the the line $\overleftrightarrow{L}$: ($x + 2y - 1 = 0$) is $\left(m = \frac{-1}{2} \right)$.

Question 2 (10 marks)

a) Define the following :
   1- The even function
   2- Midpoint of $\overline{AB}$

b) Let $f(x) = |x| - 1$, find the following :
   1- $f\,(-3)$
   2- $(D_f)\ domain\ f$
   3- determine whether f is even , odd , neither even nor odd .

Question 3 (10 marks)

a) Find the following limits (if they exist):
   1- $\lim_{x\to 2} \sqrt{4x+1}$
   2- $\lim_{x\to 2} \frac{\sqrt{x-1}-1}{x-2}$

b) Solve the following :
   1- $|x - 4\,| - 3 = -6$
   2- $\frac{x-2}{x-1} \geq 0$

c) Find the equation of the line that passes through the points $A(\,2\,,3)$, $B(5,3)$.

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